

Proof. **Theorem (Cauchy–Schwarz Inequality).** For real numbers $a_1, \dots, a_n$ and $b_1, \dots, b_n$,

$$\left(\sum_{i=1}^n a_i b_i\right)^2 \leq \left(\sum_{i=1}^n a_i^2\right) \left(\sum_{i=1}^n b_i^2\right).$$

Equality holds iff there exists a constant λ such that $a_i = \lambda b_i$ for every i (the degenerate case $b_1 = \dots = b_n = 0$ is included).

1. Construction of a non-negative quadratic form. Define

$$f(t) = \sum_{i=1}^n (a_i - t b_i)^2, \quad t \in \mathbb{R}.$$

Each summand is a square, hence $f(t) \geq 0$ for all real t .

2. Expansion of $f(t)$. Using the identity $(x - y)^2 = x^2 - 2xy + y^2$ we obtain

$$\begin{aligned} f(t) &= \sum_{i=1}^n (a_i^2 - 2a_i b_i t + t^2 b_i^2) \\ &= \underbrace{\sum_{i=1}^n b_i^2 t^2}_A - 2 \underbrace{\sum_{i=1}^n a_i b_i t}_{\frac{-B}{2}} + \underbrace{\sum_{i=1}^n a_i^2}_C. \end{aligned}$$

Thus the quadratic coefficients are

$$A = \sum_{i=1}^n b_i^2, \quad B = -2 \sum_{i=1}^n a_i b_i, \quad C = \sum_{i=1}^n a_i^2,$$

and $A \geq 0$ because it is a sum of squares.

3. Application of the quadratic discriminant theorem. The discriminant theorem states: if a quadratic $At^2 + Bt + C$ satisfies $A \geq 0$ and $At^2 + Bt + C \geq 0$ for all $t \in \mathbb{R}$, then $B^2 - 4AC \leq 0$. Since $f(t) = At^2 + Bt + C \geq 0$ for every t and $A \geq 0$, we have

$$B^2 - 4AC \leq 0.$$

Substituting the identified coefficients gives

$$\left(-2 \sum_{i=1}^n a_i b_i\right)^2 - 4 \left(\sum_{i=1}^n b_i^2\right) \left(\sum_{i=1}^n a_i^2\right) \leq 0.$$

Dividing by 4 yields the Cauchy–Schwarz inequality:

$$\boxed{\left(\sum_{i=1}^n a_i b_i\right)^2 \leq \left(\sum_{i=1}^n a_i^2\right) \left(\sum_{i=1}^n b_i^2\right)}.$$

4. Degenerate case $A = 0$. If $A = \sum_{i=1}^n b_i^2 = 0$, then each $b_i = 0$. Both sides of the inequality are zero, so the inequality holds with equality. An analogous argument applies when $C = \sum_{i=1}^n a_i^2 = 0$.

5. Equality condition. The discriminant inequality becomes an equality precisely when $B^2 - 4AC = 0$, i.e. when the quadratic $f(t)$ has a real root t_0 . Hence $f(t_0) = 0$, which means

$$\sum_{i=1}^n (a_i - t_0 b_i)^2 = 0.$$

A sum of non-negative squares can vanish only if each term is zero, so $a_i = t_0 b_i$ for all i . Conversely, if $a_i = \lambda b_i$ for some constant λ , then $f(\lambda) = 0$, the discriminant is zero, and equality holds in the Cauchy–Schwarz inequality. Thus equality occurs iff the vectors $(a_1, \dots, a_n)$ and $(b_1, \dots, b_n)$ are proportional (including the trivial case where one vector is the zero vector).

Conclusion. The Cauchy–Schwarz inequality holds for all real sequences (a_i) and (b_i) , with equality precisely in the proportional case described above.

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